Sturse.

Maternity Nursing Study Guide: Prenatal Care, Nutrition, and High-Risk Pregnancy

SECTION 01

Overview

This guide covers five lecture segments by Dr. Jonathan: prenatal care of the family, maternal and fetal nutrition, assessment of high-risk pregnancy, high-risk perinatal care with pre-existing conditions, and high-risk perinatal care with gestational conditions.

The central theme is that nurses support the whole expectant family while identifying risk early through assessment, education, and coordinated care.

SECTION 02

Key Points

Prenatal care should start as soon as the first missed period is noticed, and the lecture says it should start **before 10 weeks** so screening and baseline measurements can be done.

Prenatal care finds and treats existing or emerging maternal problems and monitors fetal growth and development.

Visit schedule: about the first three months, monthly until about week 28, every two weeks until week 36, then weekly until birth.

Nutrition before conception matters, and **folic acid** protects against neural tube defects.

Risk is assessed in four categories: biophysical, psychosocial, sociodemographic, and environmental.

Hypertensive disorders are described as the most common medical complication reported in pregnancy.

Pre-existing and gestational conditions both need specialized, interprofessional care.

SECTION 03

Nursing Concepts & Clinical Implications

Early Pregnancy Signs and Diagnosis

Signs and symptoms are grouped as **presumptive, probable, and positive**.

Pregnancy is hard to confirm before two missed periods, and many women test positive at home before their first prenatal visit.

Diagnostic accuracy matters because a wrong diagnosis can affect emotions, social life, health, or legal status.

Estimated Date of Birth

Calculated from the first day of the last normal menstrual period and the first accurate ultrasound.

Ultrasound in the first three months is usually the more accurate method.

With artificial insemination, the estimate is based on the embryo transfer date.

The **Niguel rule** (as the lecture names it) uses the woman's recall of her last normal period, assuming a cycle of about 28 days and fertilization around day 14.

The due date guides screening, growth checks, and decisions about complications.

Family and Maternal Adaptation

Each family member must adjust, and the process is shaped by culture and social trends.

Nurses care for traditional, single-parent, alternative, adoptive, and reconstituted families.

Maternal adaptation includes accepting the pregnancy, identifying with the mother role, rearranging relationships, building a bond with the unborn child, and preparing for birth.

First trimester: women are often tired and sleep more. Second trimester: fetal movement is felt and ties with the mother and other pregnant women strengthen.

Partners may feel caring, alone, or uncomfortable, may turn to friends outside the home, may take up hobbies, or may see pregnancy as proof of manhood.

Same-sex partners have similar feelings, and these can worsen when providers do not show respect.

Siblings and grandparents need recognition and inclusion in decisions.

Barriers to Prenatal Care

Poverty, lack of health insurance, lack of culturally sensitive care, and communication problems can block access.

Immigrant women may not seek prenatal care, and inadequate prenatal care is associated with low birth weight and infant mortality.

Other barriers include low motivation, inadequate finances, transportation, unpleasant past clinical experiences or trauma, clinic hours that conflict with work, child care needs, and fear that personal or cultural beliefs will not be respected.

Prenatal Care Models

Low-risk women may be seen less often and monitored remotely, which limits exposure to infectious agents.

High-risk or older women may have more frequent visits.

Group prenatal care brings together women with close due dates. Each visit starts with a one-on-one assessment, followed by group education and discussion. Families are welcome. The lecture reports high satisfaction and better perinatal outcomes, such as fewer preterm births and more breastfeeding.

The Initial Prenatal Visit

This is the longest visit and includes a full health history, psychosocial profile, physical exam, diagnostic tests, questionnaire, family background, and risk assessment.

History topics: reproductive and sexual history, medications (including herbal and illicit substances), immunization status, nutrition, family and social history, occupational history, mental health history, intimate partner violence, and a full review of systems.

Physical exam: BMI, vital signs with particular focus on blood pressure, a head-to-toe exam, and a pelvic exam. This first exam serves as the baseline for later changes.

The nurse teaches reproductive anatomy, relaxation, and breathing techniques.

Breast findings can predict breastfeeding difficulty: scars from breast reduction, implants, asymmetry, tubular shape, inadequate glandular tissue, and flat or inverted nipples.

Inverted nipples invert with gentle squeezing, unlike normal nipples, which protrude. This does not mean the woman cannot breastfeed, but it is harder.

First-Visit Laboratory and Screening Tests

Urine, cervical, and blood samples are collected. Women must be told why each test is done and may choose not to be tested.

Standard tests: blood type, Rh status, red blood cell antibody screen, complete blood count (hemoglobin and hematocrit), and rubella and varicella titers.

HIV: every woman receives risk reduction counseling and HIV testing unless she declines. A refusal must be documented.

CDC-recommended at first visit: syphilis, hepatitis B, and hepatitis C. Chlamydia and gonorrhea are tested in women under 25 and in at-risk women over 25.

Repeat in the third trimester for women at higher risk: HIV, syphilis, chlamydia, and gonorrhea. A TB skin test is done for high TB risk.

Urine: protein, glucose, nitrates, and leukocytes, plus urine culture, because pregnant women are more susceptible to UTIs.

Cervical cancer screening is done if not current. Genetic counseling is offered to couples with family history.

Follow-Up Visits and Ongoing Assessment

At each visit, ask about physical and mental health, worries, mood swings, bad dreams, and fears. These are normal in pregnancy.

Ask the partner and family how they are reacting.

Third trimester teaching includes: emergency signs of early and late labor, the labor process, fetal growth, ways to check fetal well-being, prenatal classes, and pain management.

Each visit includes blood pressure, weight, BMI trends (to ensure weight gain is adequate but not excessive), urine protein (a possible sign of high blood pressure), and edema checks.

Fundal height is the distance from the top of the uterus to the symphysis pubis. Between weeks 18 and 30, it roughly matches the weeks of gestation. The bladder should be emptied first. Steady or falling fundal height may signal **IUGR (intrauterine growth retardation)**.

Position the woman supine with a pillow under her head and a small wedge under one hip. This keeps blood pressure from dropping.

Use a paper measuring tape, document the method, and use the same method every time so measurements are reliable.

Fetal assessment includes gestational age, fetal heart tones, fetal movement, and fetal ultrasound.

Routine vaginal and rectal cultures for **Group B strep** are usually done around 36 to 37 weeks.

Genetic screening, blood tests, and amniocentesis are offered if the mother chooses.

Self-Care Education

Teach healthy nutrition, appropriate activity, good posture and body mechanics (avoid compressing vascular structures), rest, and relaxation.

Medications and herbal supplements should only be taken after checking with the provider. Illicit substances must be avoided, and immunizations kept current.

Side-lying position uses pillows between the legs, under the abdomen and hands, and under the head to promote circulation.

For leg cramps: dorsiflex the foot with the knee extended, either with a helper or by leaning on a chair.

Rh immune globulin: Rh-negative women carrying an Rh-positive baby can form antibodies against fetal red blood cells, causing hemolysis. Prophylactic Rh D immune globulin is given to Rh-negative women to prevent antibody formation.

Hygiene: hormonal changes increase sweating and odor. Watch for yeast and fungal infections, and maintain oral health because pregnant women are more prone to gingivitis.

UTI prevention: drink plenty of water, wipe front to back, avoid baths in dirty water. E. coli is one of the most common causes.

Kegel exercises strengthen the pelvic floor muscles that encircle the vaginal outlet, reducing urinary incontinence risk and aiding labor.

Clothing should be comfortable and not binding, especially shoes.

Employment is typically fine unless there are high-risk restrictions. When traveling by car, wear the seat belt correctly (below the belly and across the chest), never prop feet on the dashboard, and take frequent breaks to walk and stretch. Support stockings are advised for circulatory issues.

Begin learning about breastfeeding. Sexuality is a healthy part of pregnancy for women without high-risk issues in a monogamous relationship. Otherwise, safe sex or no sex is advised.

Teach the difference between normal discomfort and when to call the provider, and provide psychosocial support.

Maternal Age and Multifetal Pregnancy

Most U.S. pregnancies are between ages 20 and 35 and are uncomplicated.

Women younger than 20 or older than 35 may have physical and psychological adaptations that vary individually.

Multifetal pregnancy risks for mother and baby include miscarriage, hyperemesis, anemia, gestational hypertension, preeclampsia, postpartum hemorrhage, and maternal death.

Twin pregnancies often end in prematurity, IUGR or discordant growth, low and very low birth weight, congenital abnormalities, neonatal death, and cerebral palsy.

Care includes more frequent visits, tracking weight gain, monitoring nutrition, and watching for severe back pain from uterine distension.

Families may face financial strain, space needs, increased workload, and lifestyle changes. National organizations are available for support.

Perinatal Education

Goals: help families make informed, safe decisions, screen out incorrect information, and promote an empowering birth experience. Education ideally starts before pregnancy.

Early pregnancy classes: fetal growth and development, emotional and physiological changes, human sexuality, nutrition, and occupational or environmental hazards.

Mid-pregnancy classes: basic hygiene, common discomforts, breastfeeding, infant care, infant resuscitation, parenting, and labor planning.

Late pregnancy classes: managing labor discomfort, relaxation, breathing, imagery and visualization, biofeedback, and pharmacological options such as IV medications and epidurals.

Birth setting options: hospital, birth center, or home birth. High-risk pregnancies may not have all options.

Labor support: doctor, midwife, certified nurse midwife, or doula. A doula is a coach and does not deliver the baby.

A birth plan explores preferences (medications, water birth, positions) and should be flexible.

Nutrition: Assessment and Factors

Key nutrition care components: nutritional assessment (weight, height, dietary habits), diagnosis of nutrition-related problems (such as risk factors, diabetes, deficiencies, PKU, obesity), and interventions (dietary goals, weight management, varied foods, appropriate supplements, physical activity, referral to a dietitian).

Factors that influence pregnancy outcome: socioeconomic status, total nutritional intake, personal factors, environment, education, family, self-care, and overall health status and access to health services.

Ideal nutrition begins **before conception**, because the first trimester is critical for embryonic and organ development.

Folate and Neural Tube Defects

Folate is the natural form of vitamin B9. Folic acid is the form used in fortified grain products, other foods, and supplements.

Neural tube defects such as spina bifida and anencephaly result from failure of the neural tube to close, usually with none or poor folic acid intake.

The neural tube starts closing around the first month, often before pregnancy is discovered, so all women of childbearing age should take folic acid daily.

Nutrient Needs by Trimester

Needs are determined by gestational stage. The first trimester needs are only slightly increased. "Eating for two" is a misconception.

Fetal growth accelerates in the last trimester, and this is when most fetal energy and mineral stores are deposited.

Factors increasing needs: uterine-placental-fetal unit growth, total blood volume (plasma and red blood cells increase by 40 to 50 percent), breast tissue development, and about a 20 percent increase in metabolic rate.

Additional food: second trimester, one extra serving each of milk/yogurt/cheese, fruit and vegetable, and bread/cereal/rice/pasta. Third trimester, an additional one-third serving.

Quality matters: needs are for appropriate macro- and micronutrients, not excess fats, sugars, cholesterol, or fried foods.

Weight Gain

CDC recommendations: normal weight, 25 to 35 pounds. Underweight, 28 to 40 pounds. Overweight, 15 to 25 pounds. Obese, 11 to 20 pounds.

Pre-pregnancy weight relative to height is the primary consideration. Achieving a near-normal weight before pregnancy is best.

Longitudinal weight gain tracking is the best way to judge whether intake is adequate.

Underweight women gaining too little are at risk for preterm labor and IUGR.

Excessive gain increases risk for miscarriage, birth defects, stillbirth, abnormal fetal growth, preterm birth, gestational diabetes, hypertensive disorders, vacuum or forceps births, cesarean birth, surgical site infection, venous thromboembolism, and depression. Excess gain also contributes to chronic obesity and later chronic disease.

Macronutrients and Micronutrients

Protein: essential for growth and increasing demands. Primary role is amino acids for new tissue.

Fat: no more than 20 to 35 percent of daily calories. Avoid trans fatty acids, which can harm fetal development.

Carbohydrates: primary energy source. Needs increase to about 175 grams daily, constituting 45 to 64 percent of calories. Prefer whole fruits, vegetables, and whole grains. Fiber of 25 to 35 grams daily helps prevent constipation.

Micronutrients: usually met through diet except for folate and iron. Multiple micronutrient (prenatal) supplements are commonly advised, especially in the first trimester. Supplements are in addition to food, not a replacement.

Review minerals, fluids, and electrolytes in the textbook.

NURSING REMINDER

Nursing reminder: Supplements never replace a nutritious diet. Teach the pregnant woman that they are additional.

Nutrition Case Study: Ms. Brooks

A 29-year-old G1P1 at 28 weeks. Pre-pregnancy BMI of 32 and history of borderline chronic hypertension. The physician limited weight gain to 15 pounds.

At her visit she has gained 12 pounds, her blood pressure is 138/86, and her urine is negative for protein and glucose. She cries, saying she has three months to go.

Discussion points in the lecture: whether the weight gain is appropriate given BMI and hypertension, what proteinuria and glucosuria would indicate, her psychosocial state, and what the nurse should say and do.

Risk factors associated with obesity in pregnancy are an assignment for students to identify.

Other Nutrition Considerations

Avoid alcohol, limit caffeine, and limit artificial sweeteners.

Pica: craving non-food items such as baking powder, cornstarch, baking soda, laundry starch, or ice. Cravings are associated with deficiencies.

Vegetarian diets require planning to meet protein and iron needs. Gluten-free diets make healthy carbohydrates harder to obtain but not impossible.

Assess for multiparity, prior preterm or low birth weight infants, prior large-for-gestational-age infants, contraception choices, chronic illness, financial status, and nutrition knowledge.

Common nutrition-related discomforts to address: nausea and vomiting, constipation, and pyrosis.

Nutrition in Adolescents and Bariatric Surgery

Adolescents are still growing, so caloric needs differ from adult women. Plan with them, use their existing knowledge, involve a support team, and ensure access to prenatal care.

After bariatric surgery, common deficiencies include folate, vitamin B12, iron, calcium, and vitamin D.

Malabsorptive procedures (gastric bypass, biliopancreatic diversion) carry high risk for deficits. Restrictive procedures such as adjustable gastric band can also cause problems.

Screen all women with bariatric history for deficiencies throughout pregnancy, monitor weight gain, and give supplements as needed. Monitor iron closely.

Infants of mothers with bariatric surgery have increased risk of prematurity and NICU admission. Risk is greater if less than two years pass between surgery and birth.

NURSING REMINDER

Practice question from lecture: A maternity nurse should be aware of which statements are true or false: (A) oral contraceptive use may interfere with iron absorption, (B) illnesses such as phenylketonuria may require nutritional care before conception, (C) socioeconomic status and education are not relevant to the examination, (D) the only nutrition-related lab test most pregnant women need is testing for diabetes. The lecturer left the answers for class discussion.

High-Risk Pregnancy: Four Categories

Historically, risk was judged only medically. Today's approach is holistic and includes biophysical, psychosocial, sociodemographic, and environmental factors.

Biophysical: genetic disorders or defective genes, transmissible inherited disorders, chromosomal anomalies, multiple gestation, large-for-size fetus, ABO incompatibility, adolescent pregnancy, three pregnancies within two years, substance abuse history, inadequate dietary intake, excessive weight gain, hematocrit less than 33 percent, medical or obstetric illness, poorly controlled diabetes, and hypertensive disorders.

Psychosocial: disturbed interpersonal relationships, emotional distress, inadequate social support, unsafe cultural practices, substance abuse, and mental illness history.

Sociodemographic: low income, lack of prenatal care, age concerns (younger adolescent or older mother), parity, marital status, and ethnicity.

Environmental: workplace or household hazards such as chemicals, anesthetic gases, radiation, environmental pollutants that contribute to infection, and substance exposure.

Psychological effects of high-risk labeling: vulnerability, increased stress, ambivalence about the pregnancy, difficulty accomplishing parenthood tasks, or fear of motherhood.

Biophysical Assessments

Daily fetal movement count: count once daily for **60 minutes**; or two to three times daily after meals for two hours each; or 10 movements in a 12-hour period.

Ultrasound types:

> Standard (basic): most frequent; performed by ultrasound techs or trained professionals including nurses. Evaluates presentation, amniotic fluid volume, cardiac activity, placental position, fetal growth parameters, number of fetuses, and anatomic survey.
>
> Limited: targets one specific piece of information, such as fetal presentation during labor or amniotic fluid volume. Usually done by the obstetric provider.
>
> Specialized (detailed): performed when an anatomic or physiologic abnormality is suspected, after abnormal history, labs, or prior ultrasound. Done by highly trained personnel. Includes Doppler blood flow, amniotic fluid volume, and biophysical profile.

The nurse's role includes counseling and educating about the procedure.

Non-medical 3D and 4D ultrasound images are sometimes requested by families.

Ultrasound images from the textbook show the nuchal translucency and nasal bone at 12 weeks, and umbilical cord color Doppler evaluation.

MRI is not standard in pregnancy but may be indicated in high-risk cases to evaluate fetal structure and growth, placental position and density, amniotic fluid, and maternal pelvic structures.

Amniocentesis

Performed on or after 15 weeks under direct ultrasound visualization. A needle is inserted transabdominally and amniotic fluid is withdrawn. The fluid contains fetal cells and is analyzed.

Should be avoided before 13 to 14 weeks because early performance can cause miscarriage.

Indications: prenatal diagnosis of genetic disorders or congenital anomalies, especially neural tube defects; pulmonary maturity; and rarely fetal hemolytic disease.

Maternal complications: hemorrhage, infection, labor, abruptio placentae, bowel or bladder damage, and amniotic fluid embolism.

Fetal risks: death, hemorrhage, amnionitis, and needle injury. Risk is minimized by guiding the needle with ultrasound.

Women over 35 often choose it due to higher anomaly risk. It is elective.

Chorionic Villus Sampling (CVS)

Also called placental biopsy when done in the first trimester. Performed between 10 and 13 weeks.

Relatively safe, provides earlier diagnosis and faster results than amniocentesis.

Removes small tissue from the fetal portion of the placenta, not fluid.

Percutaneous Umbilical Blood Sampling (PUBS)

Provides direct access to fetal circulation in the second and third trimesters for fetal blood sampling and transfusion.

Largely replaced by CVS in many facilities because CVS is safer, easier, and faster.

Most common genetic indication: evaluation of mosaic results from amniocentesis or CVS. Also used for fetal anemia, infection, and thrombocytopenia.

Complications: bleeding from the puncture site and fetal-maternal bleeding (most common), transient fetal bradycardia, and rarely fetal loss.

Biochemical and Genetic Screening

Maternal serum alpha-fetoprotein (MSAFP): screens for neural tube defects. Detects about 85 to 92 percent of neural tube defects in almost all anencephaly cases, and about 80 to 85 percent of open neural tube and abdominal wall defects. Recommended for all pregnant women.

Multiple marker screening: detects chromosomal abnormalities and increased risk for trisomy 21.

Coombs test: detects Rh incompatibility and other maternal antibodies against fetal antigens.

Cell-free DNA screening and newer non-invasive tests can identify fetal Rh status, fetal sex, and paternally transmitted single-gene disorders. Women may choose or decline these.

Fetal Care Centers and Anomaly Diagnosis

Created to provide diagnostic and therapeutic options and support for families with a fetal anomaly diagnosis.

Interprofessional team: education, genetic counseling, social work, chaplaincy, and palliative care, because some babies will not survive after delivery.

Most centers have a coordinating nurse who helps families navigate decisions and appointments.

Antepartum Fetal Testing Principles

First and second trimester assessment focuses mainly on diagnosing fetal anomalies.

Third trimester testing determines whether the intrauterine environment continues to support fetal life.

Tests are individualized based on gestational age, pre-existing conditions, complications, and sociodemographic risks.

At least 20 percent of fetal deaths had no obvious fetal, placental, maternal, or obstetric risk identified.

No single ideal test exists for all high-risk pregnancies.

Non-Stress Test (NST)

The most widely used antepartum fetal evaluation. Based on the expectation that a normal fetus shows characteristic heart rate accelerations in response to movement, contractions, or stimulation.

Performed with the woman reclining or semi-Fowler with a slight lateral tilt to optimize uterine perfusion and prevent supine hypotension.

A Doppler transducer records fetal heart rate and a tocodynamometer detects contractions and movements. The woman may push a button to mark movement.

Usually completed in 10 to **15 minutes**; longer if the fetus must be awakened from sleep.

Opioids, propranolol, and chronic smoking can adversely affect the test.

Advantages: noninvasive, quick, inexpensive, easy to perform and interpret, and no contraindications. Disadvantages: may need to be repeated twice weekly, high false-positive rate, low false-negative rate, and less sensitive than the contraction stress test.

Reactive NST is considered normal. Non-reactive is not necessarily abnormal but requires further evaluation.

If non-reactive, testing may be extended about **20 minutes**, and vibroacoustic stimulation may be used to change fetal state. If criteria are not met after **40 minutes**, a contraction stress test may be done.

Manual stimulation and maternal orange juice are common practices, but the lecture notes no evidence that they increase fetal activity.

Contraction Stress Test (CST)

Also called the oxytocin challenge test. Performed with nipple stimulation or oxytocin to induce contractions.

Designed to identify a fetus that is stable at rest but compromised under stress. Contractions reduce uterine blood flow and placental perfusion; hypoxia produces fetal heart rate decelerations.

Provides earlier warning of fetal compromise and fewer false positives than the NST.

Limitations: cannot predict acute compromise such as umbilical cord accidents or placental abruption. More time-consuming and expensive. Invasive if oxytocin is used.

Cannot be done in women who cannot give birth vaginally at the time. Avoid in high risk for preterm labor, placenta previa, or classical cesarean incision or uterine surgery.

Usually a backup rather than a primary method.

Psychosocial Care in High-Risk Pregnancy

Women may feel stress, low self-esteem, guilt, frustration, and may avoid attachment to the fetus out of fear of loss.

Nurse roles: educate about tests and self-care, prepare for testing, counsel, assess during and after, plan interventions, support during testing, and coordinate care.

NURSING REMINDER

Nursing reminder: A reactive NST is considered normal, but a non-reactive NST requires further evaluation, not automatic diagnosis of fetal compromise.

High-Risk Pregnancy: Pre-Existing Conditions

Chronic conditions create specific maternal and fetal needs in addition to usual pregnancy needs.

Major categories: metabolic (diabetes, thyroid), cardiovascular, respiratory, gastrointestinal, integumentary, CNS, autoimmune, and substance abuse.

Diabetes in Pregnancy

The most common endocrine disorder associated with pregnancy. Key to good outcomes is strict maternal glucose control through a multidisciplinary approach, frequent blood glucose checks, and insulin administration when applicable.

Diabetes is a group of metabolic diseases with hyperglycemia from insulin defects in secretion or action, or both.

Type 1: no insulin produced. **Type 2**: insulin resistance. Other types can be infection- or drug-induced.

Gestational diabetes: any degree of glucose intolerance with onset or recognition during pregnancy in someone not already diagnosed with diabetes. It usually resolves after pregnancy, but the woman has higher risk for type 2 diabetes later.

Pre-gestational diabetes: the woman already has diabetes before pregnancy and is often insulin-dependent. Preconception counseling should address insulin changes and risks.

Maternal risks: macrosomia, hydramnios, ketoacidosis, hyperglycemia, hypoglycemia, and complications such as vascular disease, retinopathy, or nephropathy.

Fetal and neonatal risks: increased perinatal mortality, congenital malformations, respiratory distress syndrome, and intrauterine fetal demise.

Insulin changes by trimester:

> First trimester: insulin needs decrease, with nausea, vomiting, and decreased food intake possible.
>
> Second trimester: needs increase as placental hormones act as insulin antagonists.
>
> Third trimester: needs may double or quadruple, leveling off around 36 weeks.
>
> Day of birth: needs drop dramatically to near pre-pregnancy levels.
>
> Breastfeeding mothers need about 25 percent less insulin than before pregnancy. Weaning may return insulin needs toward pre-pregnancy levels.

Care management: thorough intake, physical exam, assessment of acute and chronic complications, lab results, baseline renal function, urinalysis and culture, and hemoglobin A1C. Frequent monitoring and self-monitoring. Intervene early.

Urine testing for protein, ketones, and sugar. Hospitalization may be needed for complications, with closer fetal monitoring.

Intrapartum: close glucose monitoring, possible insulin infusion, avoidance of extra IV dextrose solutions, and possible cesarean based on glucose, blood pressure, or other factors.

Postpartum: infection (such as endometritis) is more likely; hemorrhage is possible if the uterus was overdistended or overstimulated (such as with oxytocin induction). Breastfeeding is encouraged. Contraceptives need careful selection due to thromboembolic and myocardial infarction risk. Combination low-dose oral contraceptives may be prescribed to non-smoking women without vascular disease. Progestin-only oral contraceptives may be used for women with vascular disease.

Gestational Diabetes Risks and Screening

Women with gestational diabetes have fewer risks than those with pre-gestational diabetes but still have risks: preeclampsia, cesarean section, and later type 2 diabetes.

Infants of gestational diabetes mothers do not have increased birth defects, but they have higher risk for macrosomia, birth trauma, electrolyte imbalances, and neonatal hypoglycemia. Obesity contributes to congenital defects even without gestational diabetes.

Screening is typically done between 24 and 28 weeks.

Two-step method: first, a one-hour 50-gram oral glucose screen. A value below 130 is negative. A value of 130 to 140 or higher is positive and leads to a three-hour 100-gram oral glucose test. Diagnosis requires two or more values met or exceeded: fasting 95 or 105, one-hour 180 to 190, two-hour 155 to 165, or three-hour 140 to 145.

One-step method: two-hour 75-gram oral glucose tolerance test with fasting, one-hour, and two-hour values. Positive if the fasting value is 92 or higher, one-hour 180 or higher, or two-hour 153 or higher. The lecture states that one value meeting or exceeding the threshold is positive.

Both methods require fasting before testing.

Postpartum: most return to normal. Reassess at about 6 to 12 weeks. Higher risk for future gestational diabetes and type 2 diabetes.

Thyroid Disease in Pregnancy

Hyperthyroidism is rare but can occur. Graves disease causes about 90 to 95 percent of cases.

Symptoms overlap with normal pregnancy but differ in: weight loss (not normal in pregnancy), pulse over 100, and goiter. Also heat intolerance, diaphoresis, fatigue, anxiety, emotional lability, and tachycardia.

Labs: T3 and T4 elevated, TSH suppressed.

Moderate to severe hyperthyroidism must be treated. Untreated or poorly treated cases increase risk of miscarriage, preterm birth, and infertility. Women have increased risk for preeclampsia and heart failure. Most babies born to these women have normal thyroid function, though goiter may occur.

Monitor for weight gain, lethargy, decreased exercise capacity, and cold intolerance. Thyroid hormone supplements are monitored with T3, T4, and TSH levels.

The lecture omits hypothyroidism because it is very rare in pregnancy.

Phenylketonuria (PKU)

A recognized cause of intellectual disability caused by deficiency of phenylalanine hydroxylase, with toxic phenylalanine accumulation that interferes with brain development and function.

Prevention: identify women with the disorder during reproductive years. Women should be advised against breastfeeding to avoid passing the condition to the infant.

Cardiovascular Disease in Pregnancy

Normal pregnancy changes include increased intravascular volume, decreased systemic vascular resistance, cardiac output changes during labor and birth, and intravascular volume changes after birth. These can last several weeks postpartum.

A normal heart compensates. A diseased heart is hemodynamically challenged. Cardiac failure may occur during pregnancy, labor, or postpartum if changes are not tolerated.

Fever is the major cause of cardiac decompensation during pregnancy.

Cardiovascular disease complicates about 1 to 4 percent of pregnancies. Higher morbidity and mortality occur in non-white and lower-income women.

NYHA-style classes: Class 1 is asymptomatic without limitation of activity. Class 2 is symptomatic with slight limitation. Class 3 is symptomatic with marked limitation. Class 4 is symptomatic with inability to carry on activity without discomfort. Classification is done at about three months and reevaluated at seven and eight months.

Risks: increased miscarriage, preterm labor and birth, intrauterine growth restriction, and higher incidence of congenital heart lesions in offspring.

Specific conditions:

> Atrial septal defect (ASD): abnormal opening between atria, left-to-right shunt. Often undetected. Women with ASD may develop right-sided heart failure or arrhythmias, and blood clots are a concern.
>
> Ventricular septal defect (VSD): abnormal opening between ventricles. Usually corrected early. Small uncomplicated VSD has few complications. Large VSD increases risk of arrhythmias, heart failure, and pulmonary hypertension.

Thromboembolic prophylaxis is a priority.

Patent ductus arteriosus (PDA): left-to-right shunt usually corrected in infancy. Complications include endocarditis and pulmonary emboli.

Coarctation of the aorta: localized narrowing of the aorta. Hypertension in the upper extremities and hypotension in lower extremities. Asymptomatic congenital lesion. Ideally corrected before pregnancy.

Tetralogy of Fallot: most common cyanotic heart disease in pregnancy. Components: VSD, pulmonary stenosis, overriding aorta, and right ventricular hypertrophy with right-to-left shunt. Surgical repair before conception is encouraged. Poor prognosis if uncorrected and with pre-pregnancy hematocrit over 65, syncope, congestive heart failure, oxygen saturation under 90, high right ventricular pressures, or cardiomegaly. Dangerous in late third trimester and early postpartum due to reduced venous return. Use support hose, maintain blood volume, and give prophylactic antibiotics intrapartum.

Mitral valve prolapse: very common and usually benign. Leaflets prolapse into the left atrium during ventricular systole, allowing backflow. Murmur may be heard. Most are asymptomatic. Symptoms may include sharp chest pain, anxiety, palpitations, dyspnea on exertion, and syncope. Beta blockers may be used for symptoms. Thyroid checks if severe. Antibiotic prophylaxis for endocarditis is not recommended for uncomplicated cases.

Mitral stenosis: most commonly caused by rheumatic heart disease after rheumatic fever. Rheumatic fever can follow inadequately treated group A beta-hemolytic strep throat, with autoimmune damage to valves, usually the mitral valve. Stiffened valve leaflets obstruct flow. Management: reduce activity, reduce sodium, give diuretics, and assess regularly with echocardiograms.

Aortic stenosis: narrowing of the aortic valve opening, obstructing left ventricular ejection. Rarely complicates pregnancy. Management similar to mitral stenosis.

Other cardiovascular conditions named: ischemic heart disease and myocardial infarction, primary pulmonary hypertension, Marfan syndrome, infective endocarditis, Eisenmenger syndrome, peripartum cardiomyopathy, valve replacement, and heart transplant history.

Interventions:

Antepartum: minimize cardiac stress, watch for decompensation, bed rest if ordered, good nutrition with low sodium, cardiac medications (diuretics, beta blockers), and surgery if needed.

Intrapartum: frequent assessments for decompensation, arterial blood gases, electrocardiogram monitoring, promote cardiac function, and minimize anxiety.

Postpartum: monitor for decompensation, ABGs, chest pain, activity tolerance, stool softeners to avoid straining, help with infant care, and contraceptive counseling to avoid thrombosis, stroke, or heart attack risk.

Other Pre-Existing Conditions

Anemias: iron deficiency, folic acid deficiency, sickle cell hemoglobinopathy, and thalassemia.

Pulmonary: asthma (monitor exacerbations and remissions) and cystic fibrosis.

Integumentary changes: melasma, vascular spiders, palmar erythema, striae gravidarum, pruritus gravidarum, polymorphic eruption of pregnancy (papules often around the abdomen and possibly down the thighs), and intrahepatic cholestasis of

pregnancy.

Neurologic: epilepsy (women often stop medications on learning of pregnancy, which can trigger seizures; the risk of congenital anomalies from anticonvulsants must be weighed against risk of stopping them), multiple sclerosis (bed rest and steroids for acute exacerbations), Bell palsy, and myasthenia gravis.

Systemic lupus erythematosus: autoimmune disease affecting skin, joints, kidneys, lungs, and CNS. Reduce infection risk.

Myasthenia gravis: autoimmune motor disorder with muscle weakness in eyes, face, neck, limbs, and respiratory muscles. Women usually tolerate labor well but may need forceps or vacuum delivery.

Substance Abuse in Pregnancy

A dual diagnosis means substance abuse plus another psychiatric disorder.

Drug effects can stunt fetal growth, cause preterm labor, miscarriage, stillbirth, fetal alcohol syndrome, dependence, and neonatal withdrawal. Substances named: tobacco, alcohol, marijuana, opioids, cocaine, and methamphetamines.

Barriers to treatment: stigma and guilt; fear of losing custody and criminal prosecution; lack of understanding about drug effects on the baby; treatment programs that do not address pregnancy; long waiting lists; and lack of insurance. Some states treat in-utero exposure as abuse or neglect. Providers may be required to report positive results, which can damage trust.

Screening: at the first prenatal visit for past and present use, including prescription and herbal substances, using a non-judgmental approach. Toxicology may use urine, meconium, or hair.

Additional assessment: infections including HIV, hepatitis, syphilis, and other STIs. Ultrasound to determine gestational age because many women do not know they are pregnant.

Interventions: education on consequences, monitoring, treatment programs, and attention to pain management because these women may have a low pain threshold. Support bonding and address breastfeeding questions regarding exposure.

Follow-up: assess home safety, involve social services, assess support systems, and provide home care visits. If infant well-being is questionable, the case goes to Child Protective Services.

NURSING REMINDER

Nursing reminder: Substance use screening should be non-judgmental, but state reporting requirements may apply. Tell the patient what you are required to report.

Gestational Conditions: Hypertensive Disorders

Hypertensive disorders are the most common medical complication in pregnancy and a major contributor to maternal and perinatal morbidity and mortality. They complicate about 5 to 10 percent of pregnancies.

Gestational hypertension: systolic 140 or higher, diastolic 90 or higher, or both, on two or more occasions at least four hours apart, after 20 weeks, without proteinuria.

Preeclampsia: hypertension and proteinuria developing after 20 weeks in a previously normotensive woman. A vasospastic systemic disorder that can be mild or severe.

Eclampsia: preeclampsia with seizure activity or coma. Seizures can occur antepartum, intrapartum, or postpartum.

Chronic hypertension: present before pregnancy or diagnosed before 20 weeks. Can persist after delivery. No protein in urine.

Chronic hypertension with superimposed preeclampsia: sudden blood pressure increase in previously controlled hypertension, new proteinuria, or sudden sustained increase in proteinuria in a woman with proteinuria before conception or early pregnancy. Increases morbidity for mother and baby.

Some signs appear only in pregnancy and disappear after birth.

Risk factors for increased incidence and severity: multifetal pregnancy, chronic hypertension, history of preeclampsia, pre-gestational diabetes, and pre-existing thrombophilia.

Key memory point: diagnosis after 20 weeks is gestational hypertension, preeclampsia, or eclampsia. Hypertension present before or diagnosed before 20 weeks is chronic.

Preeclampsia Pathophysiology

The disorder progresses along a continuum from mild to severe and typically resolves after the placenta is expelled.

Inadequate vascular remodeling causes decreased placental perfusion and hypoxia. Placental ischemia and endothelial cell dysfunction lead to generalized vasospasm, increased peripheral resistance, and increased endothelial permeability.

Consequences of vasospasm and decreased organ perfusion: hypertension, utero-placental spasm with IUGR, glomerular damage with increased uric acid and creatinine, oliguria, cerebral spasm with headache, hyperreflexia, and seizure activity, retinal arterial spasm with blurred vision and scotomata, hyperlipidemia, and liver ischemia with elevated liver enzymes, nausea, vomiting, epigastric pain, and right upper quadrant pain.

Intravascular coagulation: hemolysis, platelet adhesion causing low platelet count with DIC, and increased factor VIII antigen.

Increased permeability and capillary leakage: proteinuria, generalized edema, pulmonary edema with dyspnea, and hemoconcentration with increased hematocrit.

Edema grading in the lecture: 1+, 2+, 3+, and 4+ pitting edema.

Assessment: blood pressure, edema, deep tendon reflexes for hyperreflexia, urine protein, and symptoms such as headache, epigastric or right upper quadrant pain, and visual disturbances.

HELLP Syndrome

Stands for hemolysis, elevated liver enzymes, and low platelets.

Diagnosed by labs. Documented as more common in White women.

Increased risk of maternal death and adverse perinatal outcomes. Undiagnosed preeclampsia is a high-risk factor.

Associated risks: pulmonary edema, acute renal failure, DIC, placental abruption, liver hemorrhage or failure, acute respiratory distress syndrome, sepsis, and stroke.

No reliable lab test screens for preeclampsia alone. Early prenatal care and risk identification are essential. Low-dose aspirin is mentioned as a possible therapy.

Preeclampsia Management

Gestational hypertension or preeclampsia without severe features: protect mother and fetus, aim for delivery close to 37 weeks, and provide outpatient or home management with close maternal and fetal assessment, frequent labs, blood pressure checks twice weekly, regular fetal movement counts, and non-stress testing.

Activity restrictions, such as bed rest, may be ordered, but the lecture notes limited evidence for benefit, so care is individualized.

Severe preeclampsia: higher risk, referral to a perinatologist (a physician specializing in the unborn fetus), antihypertensive medications, corticosteroids, and magnesium sulfate for seizure prevention and blood pressure control during labor. Also attention to future and postpartum care.

Eclampsia: immediate care with airway and safety during seizure activity, stabilization after seizure, magnesium sulfate, and checking infant status. Prevention through early identification of preeclampsia is the goal.

Chronic Hypertension Management

Chronic hypertension is associated with placental abruption, superimposed preeclampsia, heart failure, and acute kidney injury. Fetal effects include IUGR, death, and preterm birth.

Antihypertensive drugs may be discontinued in low-risk women or changed to reduce teratogenic effects.

Some antihypertensives pass into breast milk and may reduce milk production.

Hyperemesis Gravidarum

Excessive vomiting with dehydration, electrolyte imbalances, ketosis, and acetonuria. Often requires hospital admission for fluid and electrolyte replacement and observation.

Cardiovascular complications may occur, such as heart palpitations.

Clinical manifestations: inability to tolerate food and fluids, frequent vomiting, weight loss, and hair loss.

Management may require more than oral antiemetics, such as a continuous antiemetic pump (the lecture mentions ondansetron as an example of a medication used in this way) with extra doses for breakthrough nausea.

Eating bland, easily tolerated foods and avoiding strong smells may help.

Hemorrhagic Complications and Miscarriage

Maternal blood loss reduces oxygen delivery and causes risk for hypovolemia, anemia, infection, and preterm labor and birth. Fetal risks include blood loss, anemia, hypoxemia, hypoxia, anoxia, and preterm birth.

Miscarriage (spontaneous abortion) is sometimes unexplained. Risk factors discussed include cardiovascular disease, diabetes, and other disorders from earlier presentations. Management involves prenatal care, close monitoring, reducing risk factors, and psychosocial support.

Types of miscarriage: **threatened** (bleeding but not yet lost), **inevitable** (loss is certain), **incomplete** (some products expelled), **complete** (all tissue expelled), and **missed** (fetal death without expulsion).

Cervical Insufficiency

Impaired cervical strength, often diagnosed from history of preterm births or late miscarriages.

Management: **cerclage**. Vaginal cerclage closes the cervix. Abdominal cerclage places a circumferential tape around the uterine isthmus.

Follow-up: close monitoring and reporting signs of preterm labor.

Ectopic Pregnancy

A fertilized ovum implants outside the uterus and is not sustainable with life.

About 90 percent occur in the fallopian tubes, most often in the ampullary area. Other sites include the ovary, abdominal cavity, or cervix.

Clinical signs: abdominal pain, delayed menses, abnormal vaginal bleeding.

Diagnosis: quantitative HCG levels or transvaginal ultrasound.

Management: methotrexate (medical) or salpingectomy (surgical). Follow-up: weekly HCG, future fertility options, and appropriate contraception.

Molar Pregnancy

Occurs about once in every thousand pregnancies. Cause unknown. Can be complete or partial.

Clinical manifestations include blood loss or anemia.

Diagnosis: transvaginal ultrasound or serum HCG.

Management: suction curettage. Follow-up: weekly HCG monitoring and frequent pelvic exams. Emotional coping is important because the pregnancy is not sustainable.

Placenta Previa

The placenta implants in the lower uterine segment near or over the internal cervical os. Classified as complete, marginal, or low-lying, based on how much of the internal os is covered.

Affects about 1 in 200 pregnancies at term. Incidence appears to be rising, likely from more cesarean births.

Risk factors: prior cesarean, age over 35, multiparity, prior suction curettage, smoking, non-white race, maternal cocaine use, pregnancies from reproductive technology, multiple gestations, male fetuses, and prior placenta previa.

Classic presentation: painless, bright red vaginal bleeding in the second or third trimester. About 90 percent have at least one bleeding episode. Diagnosis is now usually by ultrasound before significant bleeding.

Vital signs may be normal despite heavy loss because pregnant women can lose about 40 percent of blood without showing shock. Urine output and clinical presentation are better indicators.

Fetal heart rate is normal unless major placental detachment occurs. The uterus is soft, relaxed, and non-tender. Fundal height is often larger than expected. Malpresentation such as breech is common.

Placental Abruption

About one in one thousand births. Can be mild, moderate, or severe.

Manifestations: vaginal bleeding, abdominal pain, uterine tenderness, and contractions.

Diagnosis is primarily clinical. Close monitoring is essential because abruption can occur at any time.

Moderate to severe bleeding: check blood work and vital signs, monitor hemodynamic changes, use continuous fetal monitoring, provide fluid resuscitation as needed. Vaginal birth is preferred, but cesarean may be necessary. The goal is to prevent major blood loss.

Classification: retroplacental (blood behind placenta), subchorionic (bleeding beneath chorion), and preplacental (blood anterior to placenta, within amnion and chorion).

Other placental abnormalities shown in the textbook: velamentous cord insertion and battledore placenta.

Disseminated Intravascular Coagulation (DIC)

Normal hemostasis balances clotting and fibrinolysis. Clotting forms a platelet plug and fibrin to stop bleeding from injured vessels.

DIC is a pathologic diffuse clotting process causing widespread internal and external bleeding.

Usually triggered by severe preeclampsia, HELLP syndrome, or gram-negative sepsis.

Management: find and treat the cause, assess for bleeding, and stop the bleeding.

Infections and Surgery in Pregnancy

Infections requiring careful monitoring: sexually transmitted infections, urinary tract infections, asymptomatic bacteriuria, cystitis, and pyelonephritis. Patients need education on managing and preventing infection.

Surgery may be needed for appendicitis, cholelithiasis, or cholecystitis during pregnancy. Untreated conditions carry risks that may outweigh surgical risks. Perinatal and obstetric care is coordinated closely.

Trauma in Pregnancy

Trauma is often underreported. Falls and motor vehicle accidents are common. Prevent through seat belt use, proper footwear, and caution with activities such as hiking as weight and balance change.

Outcome depends on gestational age, type and severity of trauma, and physiological disruptions.

Maternal physiology: the bladder and uterus are abdominal, tolerance for hypoxia is decreased, and significant blood loss may be less obvious on vital signs.

Fetal assessment depends on maternal arterial pressure and electronic fetal monitoring.

Mechanisms: blunt trauma, penetrating trauma, and thoracic trauma are the more severe types.

Management: immediate stabilization of mother and baby, which may include CPR, a primary survey (life-threatening issues), and a secondary survey (all other system assessment). Electronic fetal monitoring, checking for fetal-maternal

hemorrhage, ultrasound, and assessment of radiation exposure. Perimortem cesarean delivery may be needed if the mother has died.

NURSING REMINDER

Nursing reminder: Pregnant women can lose about 40 percent of blood volume before showing signs of shock. Do not rely on vital signs alone to judge hemorrhage.

SECTION 04

Core Definitions

Presumptive signs: subjective changes reported by the woman that suggest pregnancy.

Probable signs: objective findings that suggest pregnancy but can have other causes.

Positive signs: definitive evidence of pregnancy.

Fundal height: distance from the top of the uterus to the symphysis pubis, used to assess growth.

Group prenatal care: a model where women with close due dates receive assessments individually and education in a group setting.

Gestational diabetes: glucose intolerance first recognized during pregnancy in a woman without prior diabetes.

Pre-gestational diabetes: diabetes diagnosed before pregnancy.

Non-stress test (NST): monitoring of fetal heart rate response to movement or contractions to assess fetal well-being.

Contraction stress test (CST): fetal heart rate monitoring during induced contractions to detect compromise under stress.

Biophysical profile: a specialized ultrasound-based assessment of fetal well-being.

Gestational hypertension: new hypertension after 20 weeks without proteinuria.

Preeclampsia: new hypertension and proteinuria after 20 weeks in a previously normotensive woman.

Eclampsia: preeclampsia with seizures or coma.

HELLP syndrome: hemolysis, elevated liver enzymes, and low platelets, a severe complication of preeclampsia.

Placenta previa: placenta implanted in the lower uterine segment near or over the internal cervical os.

Placental abruption: separation of the placenta from the uterine wall before birth.

Ectopic pregnancy: implantation of a fertilized ovum outside the uterus.

SECTION 05

NCLEX-Style Focus Areas

Recognize why prenatal care should begin early, and what baseline measurements and screenings happen at the first visit.

Know the standard first-visit labs, including Rh status and the reason for Rh immune globulin in Rh-negative women.

Interpret fundal height findings and know that steady or falling measurements may indicate IUGR.

Know which substances and conditions require the nurse to use non-judgmental screening and report requirements.

Compare the four risk categories: biophysical, psychosocial, sociodemographic, and environmental.

Know when a non-stress test is reactive versus non-reactive, and when a contraction stress test is contraindicated (preterm labor risk, placenta previa, classical incision).

Differentiate amniocentesis, CVS, and PUBS by timing and purpose.

Know the blood pressure and timing thresholds that distinguish gestational hypertension, preeclampsia, eclampsia, and chronic hypertension.

Recognize HELLP syndrome and DIC as complications of severe preeclampsia.

Know the difference between gestational and pre-gestational diabetes, insulin changes by trimester, and neonatal hypoglycemia risk in infants of diabetic mothers.

Recognize the signs of hyperthyroidism versus normal pregnancy changes, and the lab pattern: elevated T3 and T4, suppressed TSH.

Identify the classic presentation of placenta previa (painless bright red bleeding) versus abruption (pain, tenderness, and contractions).

Know the types of miscarriage and the three signs of ectopic pregnancy named in the lecture.

Recognize the nursing priorities for cardiac disease in pregnancy: minimize stress on the heart, monitor for decompensation, and watch for fever as a trigger.

Know the nutritional deficiencies at risk after bariatric surgery and the weight gain ranges by pre-pregnancy BMI category.